

DCME

Does the Los Angeles Clippers Basketball Team Have a Home Court Advantage?

Heather LaRange

INTUIT

LA Clippers

Location: Los Angeles

Founded: 1984

Home Court: Intuit Dome

- Holds 18,000 fans

Accomplishments:

- Qualified for postseason 19 times
- 2 division titles 2013 and 2014

2024–2025 Season Ranking: 5th out of 15 in Western Conference



The Wall

- Built before the 2024–2025 season
- Largest fan section spanning 51 rows of the most dedicated, loyal fans
- Fans are equipped with props to distract the players, uniquely designed for different opponents
- It is behind the opposing team's hoop in the first half
- Fans are screened to ensure they are the most hardcore



The Wall



The Study

Our retrospective study will include the population of the Los Angeles Clippers' regular season schedule home and away games. We will pull data from our sampling frame of a list of all 82 Clippers regular-season games in the 2024–2025 season. There are two variables of interest: game location and game outcome. Both of these are categorical variables as game locations are “home” or “away” and game outcomes are “win” or “loss”.

2024-2025 Regular Season

Overall Record: 50-32
Home Record: 30-11
Away Record: 20-21

DATE	LOCATION	RESULT	W-L	DATE	LOCATION	RESULT	W-L	DATE	LOCATION	RESULT	W-L
Thurs, Oct 23	HOME	L	0-1	Sat, Jan 4	HOME	W	20-15	Tue, Mar 18	HOME	W	39-30
Sun, Oct 26	AWAY	W	1-1	Mon, Jan 6	AWAY	L	20-16	Fri, Mar 21	HOME	W	40-30
Mon, Oct 27	AWAY	W	2-1	Wed, Jan 8	AWAY	L	20-17	Sun, Mar 23	HOME	L	40-31
Thurs, Oct 30	HOME	L	2-2	Mon, Jan 13	HOME	W	21-17	Wed, Mar 26	AWAY	W	41-31
Fri, Oct 31	HOME	L	2-3	Wed, Jan 15	HOME	W	22-17	Fri, Mar 28	AWAY	W	42-31
Sun, Nov 2	HOME	L	2-4	Thurs, Jan 16	AWAY	W	23-17	Sun, Mar 30	AWAY	L	42-32
Tue, Nov 4	HOME	W	3-4	Sun, Jan 19	HOME	W	24-17	Mon, Mar 31	AWAY	W	43-32
Thurs, Nov 6	HOME	W	4-4	Mon, Jan 20	HOME	L	24-18	Wed, Apr 2	HOME	W	44-32
Sat, Nov 8	AWAY	W	5-4	Wed, Jan 22	HOME	L	24-19	Fri, Apr 4	HOME	W	45-32
Sun, Nov 9	HOME	W	6-4	Thurs, Jan 23	HOME	W	25-19	Sat, Apr 5	HOME	W	46-32
Tue, Nov 11	AWAY	L	6-5	Sat, Jan 25	HOME	W	26-19	Tue, Apr 8	HOME	W	47-32
Thurs, Nov 13	AWAY	L	6-6	Mon, Jan 27	AWAY	L	26-20	Wed, Apr 9	HOME	W	48-32
Sat, Nov 15	AWAY	L	6-7	Wed, Jan 29	AWAY	W	27-20	Fri, Apr 11	AWAY	W	49-32
Mon, Nov 17	HOME	W	7-7	Fri, Jan 31	AWAY	W	28-20	Sun, Apr 13	AWAY	W	50-32
Tue, Nov 18	HOME	W	8-7	Sun, Feb 2	AWAY	L	28-21				
Thurs, Nov 20	HOME	W	9-7	Tue, Feb 4	HOME	L	28-22				
Sat, Nov 22	HOME	W	10-7	Thu, Feb 6	HOME	L	28-23				
Mon, Nov 24	AWAY	W	11-7	Sat, Feb 8	HOME	W	29-23				
Tue, Nov 25	AWAY	L	11-8	Wed, Feb 12	HOME	W	30-23				
Thurs, Nov 27	AWAY	W	12-8	Thu, Feb 13	AWAY	W	31-23				
Sat, Nov 29	AWAY	L	12-9	Thu, Feb 20	AWAY	L	31-24				
Mon, Dec 1	HOME	W	13-9	Sun, Feb 23	AWAY	L	31-25				
Wed, Dec 3	HOME	W	14-9	Mon, Feb 24	AWAY	L	31-26				
Thurs, Dec 4	HOME	L	14-10	Wed, Feb 26	AWAY	W	32-26				
Mon, Dec 8	HOME	L	14-11	Fri, Feb 28	AWAY	L	32-27				
Sat, Dec 13	AWAY	L	14-12	Sun, Mar 2	AWAY	L	32-28				
Tue, Dec 16	HOME	W	15-12	Tue, Mar 4	AWAY	L	32-29				
Fri, Dec 19	AWAY	W	16-12	Wed, Mar 5	HOME	W	33-29				
Sun, Dec 21	AWAY	L	16-13	Fri, Mar 7	HOME	W	34-29				
Tue, Dec 23	AWAY	W	17-13	Sun, Mar 9	HOME	W	35-29				
Sat, Dec 27	HOME	W	18-13	Tue, Mar 11	AWAY	L	35-30				
Tue, Dec 30	AWAY	W	19-13	Wed, Mar 12	AWAY	W	36-30				
Wed, Dec 31	AWAY	L	19-14	Fri, Mar 14	AWAY	W	37-30				
Thurs, Jan 2	AWAY	L	19-15	Sun, Mar 16	HOME	W	38-30				

Two-Proportion Z-Test

$$H_0: p_1 - p_2 = 0$$

$$H_A: p_1 - p_2 > 0$$

Where p_1 = the true proportion of Los Angeles Clippers home games won

p_2 = the true proportion of Los Angeles Clippers away games won

$$\alpha = 0.05$$

Conditions

$$\begin{aligned}\hat{p}_c &= (x_1 + x_2)/(n_1 + n_2) \\ &= (30 + 20)/(41 + 41) = 0.61\end{aligned}$$

Independence Assumption : we will assume each regular season game is independent of the next

Independent Groups Assumption : we will assume regular season home games are independent of regular season away games

Sample Size Assumption

Success/Failure Condition:

$$n_1(\hat{p}_c) = 41(0.61) = 25.01$$

$$n_2(\hat{p}_c) = 41(0.61) = 25.01$$

$$n_1(1-\hat{p}_c) = 41(0.39) = 15.99$$

$$n_2(1-\hat{p}_c) = 41(0.39) = 15.99$$

We expect at least
10 successes and
10 failures in each
group.

Okay to use Normal model and two-proportion z-test.

Two-Proportion Z-Test

$$n_1 = 41$$

$$x_1 = 30$$

$$\hat{p}_1 = 30/41 = 0.732$$

$$\hat{p}_c = 0.61$$

$$\hat{p}_1 - \hat{p}_2 = 0.732 - 0.488 = 0.244$$

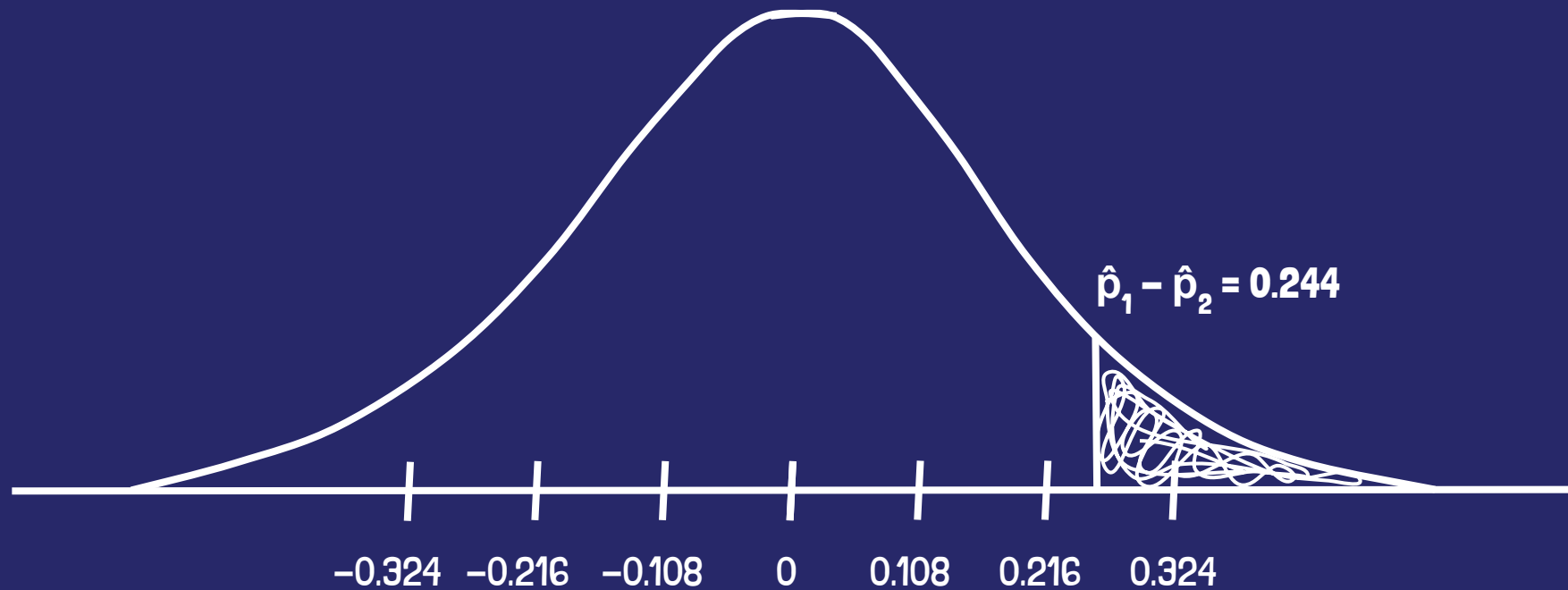
$$\begin{aligned} SE_c(\hat{p}_1 - \hat{p}_2) &= \sqrt{[(\hat{p}_c(1-\hat{p}_c))/n_1] + [(\hat{p}_c(1-\hat{p}_c))/n_1]} \\ &= \sqrt{[(0.61(1-0.61))/41] + [(0.61(1-0.61))/41]} \\ &= 0.108 \end{aligned}$$

$$n_2 = 41$$

$$x_2 = 20$$

$$\hat{p}_2 = 20/41 = 0.488$$

$N(0, 0.108)$



Two-Proportion Z-Test

$$z = (\hat{p}_1 - \hat{p}_2 - 0) / \text{SE}_c(\hat{p}_1 - \hat{p}_2) = 0.244 / 0.108 = 2.26$$

$$\begin{aligned} \text{p-value} &= P(\hat{p}_1 - \hat{p}_2 > 0) \\ &= P(z > 2.26) \\ &= 0.0119 \end{aligned}$$

Since the p-value of 0.0119 is below 0.05, we reject the null hypothesis. There is strong enough evidence to say that the Los Angeles Clippers do have a home court advantage.

Conclusion

Based on the hypothesis test we can conclude that efforts the Clippers have made to make their home court give them an advantage against their opponents appear to have been successful. In a future study we could test the Clippers home win percentage against previous seasons, to see if their new stadium has made a significant difference from their previous stadium.